Enova International: Fraud Detection in FinTech

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Agenda

01 Problem Statement

02 Data & Methodology

03 Results

04 Conclusion

Problem Statement

ENOVA

Company

ENOVA is a leading **digital loan lending** company founded in 2004

Challenge

Limited operations capacity to manually review loan applications

Approach

Reliance on **machine learning algorithms** to flag risky applicants

Project Goal

- Feature engineering
- **Flag the 5% riskiest applicants** (alignment with view capacity)

Data Overview

1/ Internal Data Sources

Complete People
Basic customer information

Customer Changes
Applicant information updates

Bank Account History
Bank information updates

2/ External Data Sources

Iovation
Device intelligence data

Emailage
Email-based risk signals

RiskView
Applicant behavioral credit data

3/ Training Dataset

- **Applicant-level internal and external features**
- **Historically labeled fraud outcomes**

Evaluate engineered feature performance
Train models

Methodology

Baseline Model

Trained on existing features

Feature Engineering

Create new, meaningful features

Advanced Model

Trained on existing and new features

CatBoost

Handle **mixed and high-cardinality categorical features**

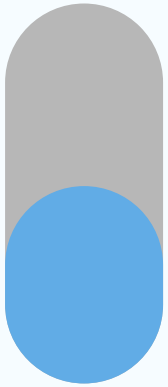
Perform well in **imbalanced fraud detection** settings

Treat **missing values** as informative

Apply ordered boosting with regularization to **prevent overfitting**

Feature Engineering

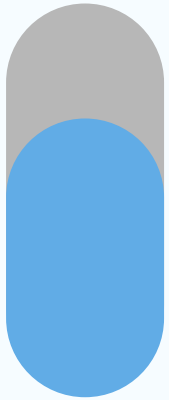
25%



Cross-Features Logical Patterns

Combine signals when multiple risk conditions co-occur

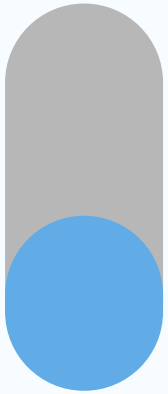
45%



Behavioral Indicators

Model realistic fraud behaviors rather than static attributes

20%

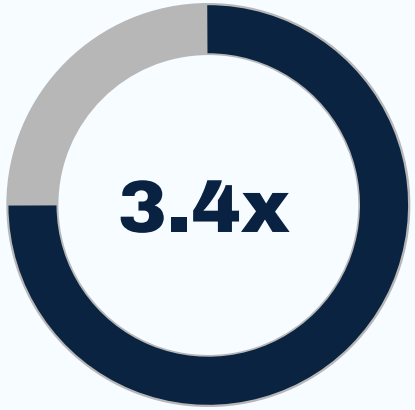


Time-Based Features

Time-based windows for suspicious patterns

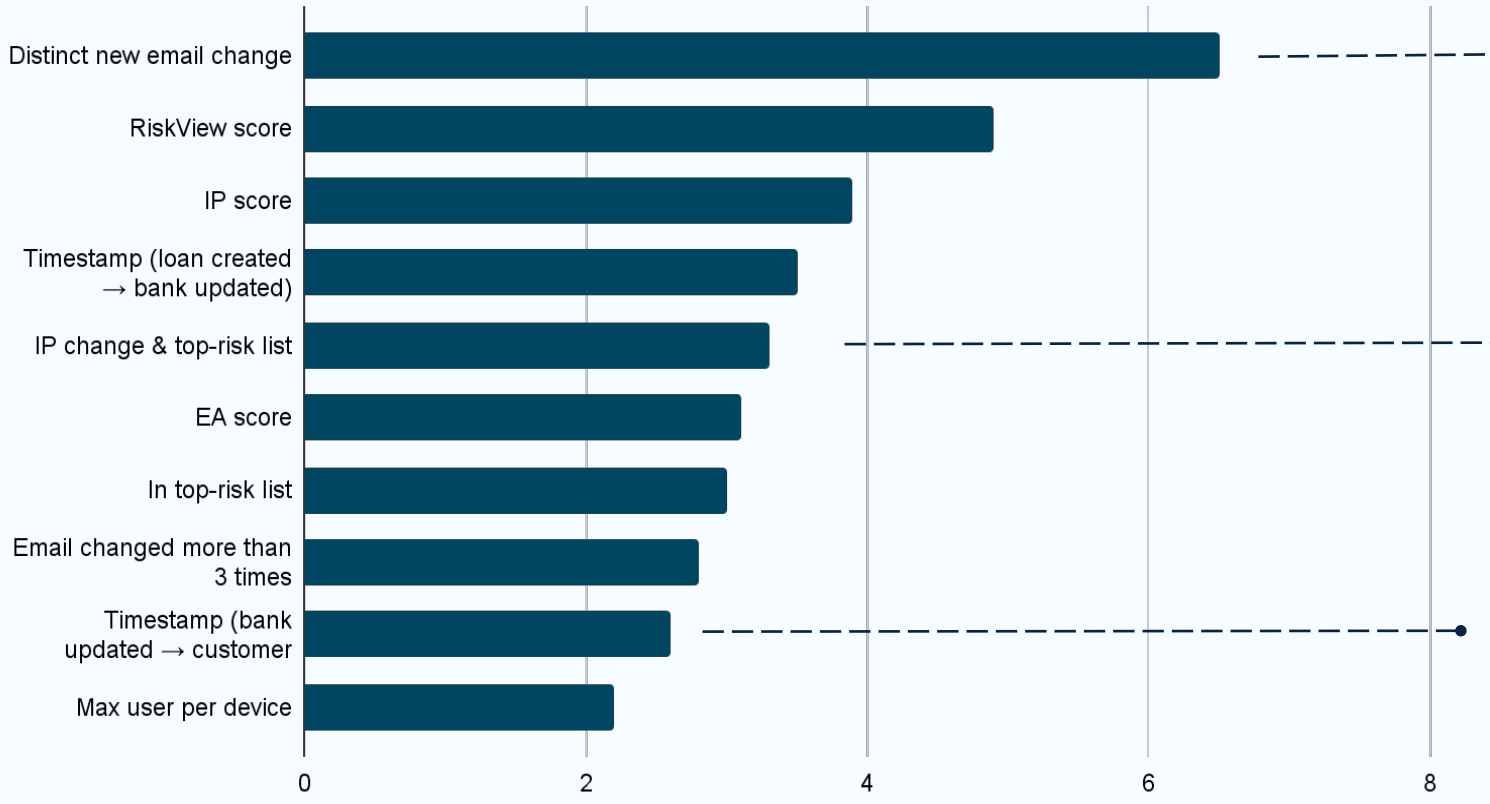
Model Performance Results

Dataset	Model	Recall	Accuracy
Train	Baseline	13.38%	98.27%
	Advanced	31.21%	98.27%
Validation	Baseline	9.62%	97.74%
	Advanced	28.85%	98.22%
Test	**Baseline**	**7.96%**	97.70%
	Advanced	**26.92%**	98.18%



Improvement in recall without sacrificing accuracy

Feature Importance (Top 10)



Behavioral Indicators: Strongly predictive because real fraud is driven by instability in email, IP, device, and account behaviors

Cross-Features: Capture high-risk behaviors that emerge when multiple signals align, boosting predictive power

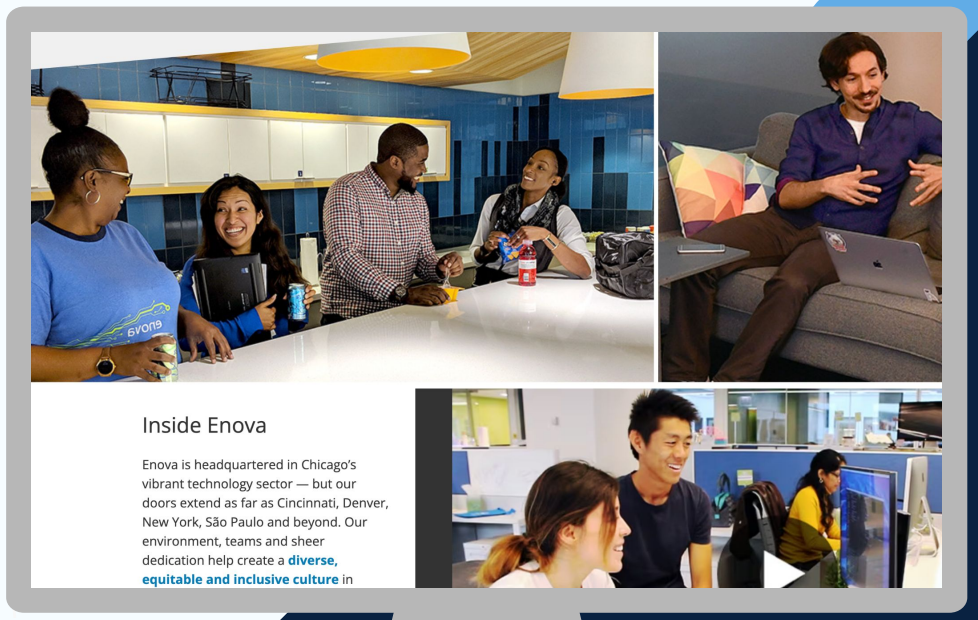
Time-Based Features: Reveal irregular event timing, a hallmark of manipulated or synthetic identities

Business Impact

✔ Improved baseline recall score by 19%

✔ Flagged 5% riskiest applicants without requiring more review capacity

✔ Reduced reliance on static vendors by introducing dynamic behavioral and temporal features



Future Work

Business intuition for **prioritizing recall**

Stronger Regularization and Model Interpretability

Model Robustness and **Real-World Interpretation**

Integrating Optimization for Business Profit Maximization

Our project successfully delivers operationally feasible fraud indicators that enhance Enova's fraud detection capabilities

Thank you for your time!

Questions?